Starry Night Project Using Arduino Nano and Optical Fibres

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Abstract

This project replicates a starry night effect using an Arduino Nano and WS2812B LEDs. A 64-bit Linear Feedback Shift Register (LFSR) generates random brightness and timing for a realistic twinkling pattern. Optical fibres distribute light from LEDs through a cardboard surface to simulate glowing stars.

Components Used

- Arduino Nano
- WS2812B LED strip (14 LEDs)
- 12×12 cm cardboard sheet with additional 2-inch margin
- Small cardboard pieces (for back support)
- Zero board
- 3-pin JST male-female connector
- 470 Ω resistor
- Optical fibres
- 2-part epoxy adhesive
- USB cable
- Paper knife and compass

Construction Procedure

1. A 12×12 cm cardboard sheet was prepared, with an additional 2-inch margin on one side for mounting the zero board.
2. A 10×10 cm central grid was marked for drilling holes that would act as star openings.

3. The holes were made using a compass, and the sheet edges were trimmed neatly with a paper knife.
4. Small rectangular cardboard pieces were glued at the back on all four sides to act as support stands.
5. The WS2812B LED strips were first glued on the cardboard supports using epoxy.
6. Optical fibres were then glued onto the LEDs using epoxy. Each LED was connected to about 20–24 fibres, allowing one LED to light multiple star holes.
7. The cardboard supports (with LED strips and fibres attached) were then glued to the back side of the main cardboard sheet.
8. After this, the free ends of the optical fibres were glued into the holes on the front side of the main sheet.
9. The fibre lengths were decided based on the distance from the LEDs:
    - Short fibres for nearby holes
    - Long fibres for farther holes
10. Each hole contained 2–4 fibres depending on its size and desired brightness.
11. All glueing was done from the back side so that the front surface remained smooth and clean, giving a clear starry night view.
12. Finally, a zero board was made containing the Arduino Nano, 3-pin JST connector, and 470 Ω current limiting resistor. The Arduino was soldered on this board and mounted on the extra 2-inch margin.
13. The JST connector was used to connect the LED strip to the zero board for easy detachment.
14. A hole was drilled near the zero board to pass the USB cable for Arduino power and programming.

Circuit Connections

- LED strip VCC → 5V (Arduino Nano)
- LED strip GND → GND (Arduino Nano)
- LED strip DIN → D4 (via 470 Ω resistor)

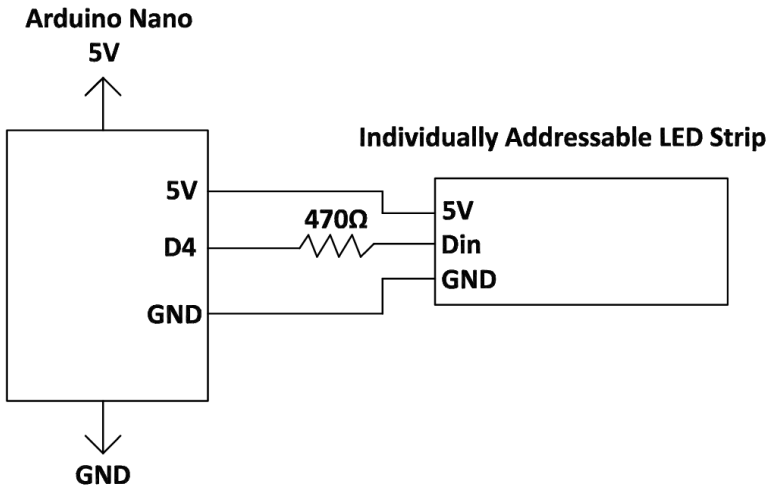


Figure 1: Block Diagram of the Starry Night Project

Arduino Code

```
#include <FastLED.h>
#include <stdint.h>
#define NUM_LEDS 14
#define LED_PIN 4
uint64_t lfsr; uint8_t bit;
CRGB leds[NUM_LEDS];

void setup() {
lfsr = 0;
for (int i = 0; i < 64; i++) {
int noise = analogRead(A0);
lfsr = (lfsr << 1) | (noise & 1);
delay(20);
}
if (lfsr == 0) lfsr = 0xACE1uLL << 48 | 0x1234uLL;
FastLED.addLeds<WS2812B, LED_PIN, GRB>(leds, NUM_LEDS);
FastLED.setMaxPowerInVoltsAndMilliamps(5, 500);
FastLED.setBrightness(255);
}

void loop() {
for(int i = 0; i < NUM_LEDS; i++){
leds[i] = CRGB(255,255,255);
bit = ((lfsr>>0)^(lfsr>>1)^(lfsr>>3)^(lfsr>>4)) & 1;
lfsr = (lfsr >> 1) | ((uint64_t)bit << 63);
leds[i].nscale8(lfsr & 0xFF);
FastLED.show();
delay(lfsr & 0x3F);
}
}
```

Listing 1: Arduino Code for Starry Night Project

Results and Observations

The optical fibres successfully transmitted light from the LEDs to the front surface, creating a natural starry night effect. Each LED was glued with around 20–24 fibres.

Because so many fibres are attached to a single RGB LED, different fibres captured different color components — red, green, or blue — even when the LED was programmed to emit white light. This accidental color separation resulted in a mix of white, red, green, and blue stars, which enhanced the overall visual beauty of the project and made the night sky look more dynamic and realistic.

The LED strips mounted on the back support pieces provided stable alignment and improved light distribution, while the 3-pin JST connector and back-side wiring kept the setup clean and serviceable.

Photographs

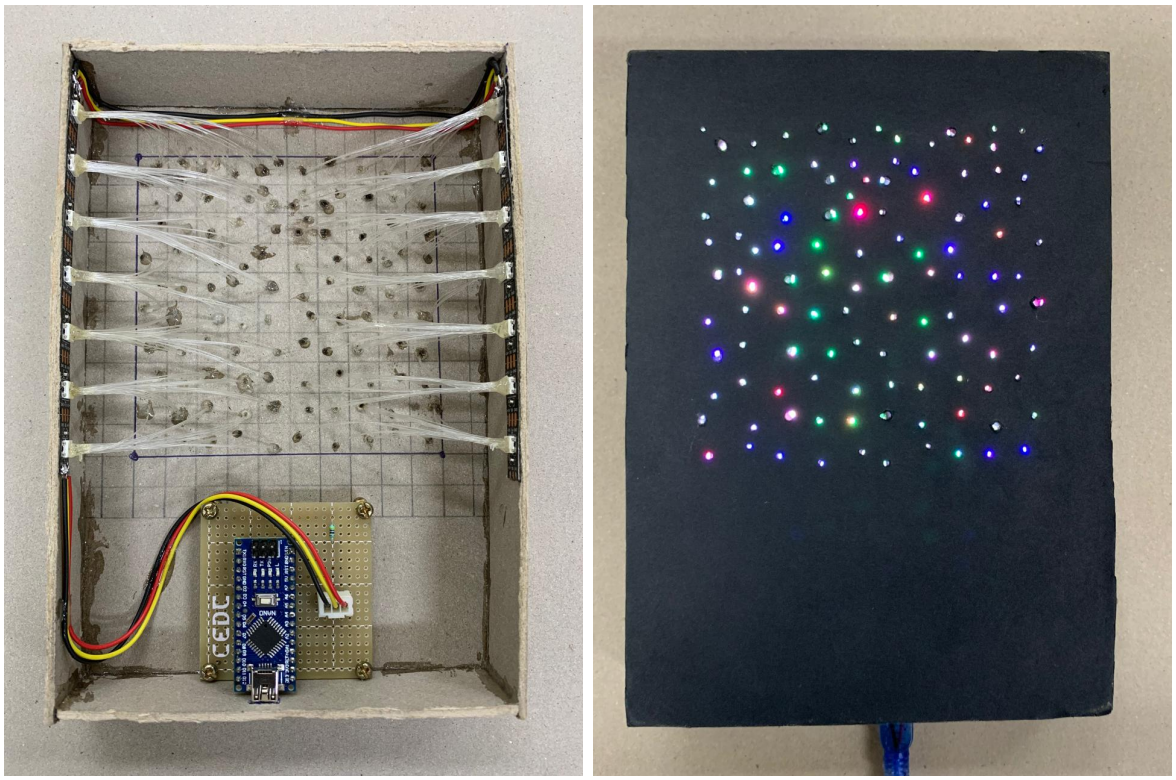


Figure 2: Left: Internal setup showing Arduino Nano, zero board, LED strips, and fibres. Right: Final glowing starry night effect.

Conclusion

The Starry Night project successfully combines electronics and optics to produce a realistic star-field effect. The 64-bit LFSR algorithm added randomness for natural twinkling, and the RGB colour separation through optical fibres unexpectedly enhanced visual richness. The use of a simple cardboard structure, epoxy, and fibre optics demonstrates how inexpensive materials can yield beautiful artistic results. Future enhancements could include RGB animation effects or sound-reactive lighting.